Lab14.4_NER_spaCy_HF_N.Rukmini_2403A52402

STEP 1 — Install Required Libraries

```
!pip install spacy transformers torch pandas datasets
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
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Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
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Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
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Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
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Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
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Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
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Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: bliss<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->sp
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Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=
```

STEP 2 — Import Libraries

```
import spacy                        # NLP library for tokenization & NER
from spacy import displacy          # Visualization of entities
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification  # HF models
import torch                        # Deep learning framework
import pandas as pd                 # Data handling
```

STEP 3 — Load spaCy Model

```
nlp = spacy.load("en_core_web_sm")
```

What is a pipeline?

A pipeline in spaCy is a sequence of processing components applied to text.

spaCy pipeline components:

Tokenizer – Splits text into tokens

Tagger – Assigns POS tags

Parser – Finds syntactic relations

NER – Detects named entities

STEP 4 — Sample Text Data

```
sentences = [
    "Apple announced new AI features at WWDC 2024 in California.",
    "Virat Kohli scored a century in the match against Australia in Mumbai.",
    "Prime Minister Narendra Modi visited the United States last week.",
    "Microsoft acquired Activision Blizzard for $69 billion.",
    "The Chandrayaan-3 mission was successfully launched by ISRO."
]
```

STEP 5 — Apply spaCy NER

```
print("---- spaCy NER Results ----\n")

for sentence in sentences:
    doc = nlp(sentence)
    print(f"Sentence: {sentence}")
    for ent in doc.ents:
        print(f"Entity: {ent.text} | Label: {ent.label_}")
    print("-" * 50)
```

```
---- spaCy NER Results ----

Sentence: Apple announced new AI features at WWDC 2024 in California.
Entity: Apple | Label: ORG
Entity: AI | Label: GPE
Entity: 2024 | Label: DATE
Entity: California | Label: GPE
--------------------------------------------------
Sentence: Virat Kohli scored a century in the match against Australia in Mumbai.
Entity: Virat Kohli | Label: PERSON
Entity: a century | Label: DATE
Entity: Australia | Label: GPE
Entity: Mumbai | Label: GPE
--------------------------------------------------
Sentence: Prime Minister Narendra Modi visited the United States last week.
Entity: Narendra Modi | Label: PERSON
Entity: the United States | Label: GPE
Entity: last week | Label: DATE
--------------------------------------------------
Sentence: Microsoft acquired Activision Blizzard for $69 billion.
Entity: Microsoft | Label: ORG
Entity: Activision Blizzard | Label: PERSON
Entity: $69 billion | Label: MONEY
--------------------------------------------------
Sentence: The Chandrayaan-3 mission was successfully launched by ISRO.
Entity: ISRO | Label: ORG
--------------------------------------------------
```

STEP 6 — Store spaCy Results in Table

```
spacy_results = []

for sentence in sentences:
    doc = nlp(sentence)
    for ent in doc.ents:
        spacy_results.append([sentence, ent.text, ent.label_])

df_spacy = pd.DataFrame(spacy_results, columns=["Sentence", "Entity", "Label"])
df_spacy
```

	Sentence	Entity	Label
0	Apple announced new AI features at WWDC 2024 i...	Apple	ORG
1	Apple announced new AI features at WWDC 2024 i...	AI	GPE
2	Apple announced new AI features at WWDC 2024 i...	2024	DATE
3	Apple announced new AI features at WWDC 2024 i...	California	GPE
4	Virat Kohli scored a century in the match agai...	Virat Kohli	PERSON
5	Virat Kohli scored a century in the match agai...	a century	DATE
6	Virat Kohli scored a century in the match agai...	Australia	GPE
7	Virat Kohli scored a century in the match agai...	Mumbai	GPE
8	Prime Minister Narendra Modi visited the Unite...	Narendra Modi	PERSON
9	Prime Minister Narendra Modi visited the Unite...	the United States	GPE
10	Prime Minister Narendra Modi visited the Unite...	last week	DATE
11	Microsoft acquired Activision Blizzard for $69...	Microsoft	ORG
12	Microsoft acquired Activision Blizzard for $69...	Activision Blizzard	PERSON
13	Microsoft acquired Activision Blizzard for $69...	$69 billion	MONEY
14	The Chandrayaan-3 mission was successfully lau...	ISRO	ORG

STEP 7 — Visualize Entities (Optional)

```
doc = nlp(sentences[0])
displacy.render(doc, style="ent", jupyter=True)
```

Apple ORG announced new AI GPE features at WWDC 2024 DATE in California GPE .

Explanation:

Different entity types (PERSON, ORG, GPE, MONEY, DATE) appear in different colors.

Labels appear above highlighted text.

STEP 8 — Load Hugging Face NER Model

```
model_name = "dbmdz/bert-large-cased-finetuned-conll03-english"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)

ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%  998/998 [00:00<00:00, 62.1kB/s]
tokenizer_config.json: 100%  60.0/60.0 [00:00<00:00, 2.56kB/s]
vocab.txt: 213k/? [00:00<00:00, 4.32MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
model.safetensors: 100%  1.33G/1.33G [00:29<00:00, 66.0MB/s]
Loading weights: 100%  391/391 [00:01<00:00, 381.82it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                       | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.weight | UNEXPECTED |  |
bert.pooler.dense.bias   | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

What is Token Classification?

Token classification assigns a label to each token (e.g., PERSON, ORG, LOC).

Why Transformers understand context better?

Models like Hugging Face Transformers:

Use self-attention

Consider full sentence context

Produce contextual embeddings

Capture long-range dependencies

STEP 9 — Apply Transformer NER

```
print("---- Hugging Face NER Results ----\n")

for sentence in sentences:
    results = ner_pipeline(sentence)
    print(f"Sentence: {sentence}")
    for r in results:
        print(f"Word: {r['word']} | Label: {r['entity']} | Score: {round(r['score'],4)}")
    print("-" * 50)
```

```
---- Hugging Face NER Results ----

Sentence: Apple announced new AI features at WWDC 2024 in California.
Word: Apple | Label: I-ORG | Score: 0.9977999925613403
Word: W | Label: I-MISC | Score: 0.9348000288009644
Word: ##WD | Label: I-MISC | Score: 0.9257000088691711
Word: ##C | Label: I-MISC | Score: 0.9882000088691711
Word: California | Label: I-LOC | Score: 0.9995999932289124
--------------------------------------------------
Sentence: Virat Kohli scored a century in the match against Australia in Mumbai.
Word: V | Label: I-PER | Score: 0.9995999932289124
Word: ##ira | Label: I-PER | Score: 0.9962999820709229
Word: ##t | Label: I-PER | Score: 0.9991000294685364
Word: Ko | Label: I-PER | Score: 0.9994999766349792
Word: ##hl | Label: I-PER | Score: 0.9907000064849854
Word: ##i | Label: I-PER | Score: 0.994700014591217
Word: Australia | Label: I-LOC | Score: 0.9997000098228455
Word: Mumbai | Label: I-LOC | Score: 0.9990000128746033
--------------------------------------------------
Sentence: Prime Minister Narendra Modi visited the United States last week.
Word: Na | Label: I-PER | Score: 0.9991999864578247
Word: ##ren | Label: I-PER | Score: 0.9925000071525574
Word: ##dra | Label: I-PER | Score: 0.9983999729156494
Word: Mo | Label: I-PER | Score: 0.9988999962806702
Word: ##di | Label: I-PER | Score: 0.9837999939918518
Word: United | Label: I-LOC | Score: 0.9997000098228455
Word: States | Label: I-LOC | Score: 0.9998000264167786
--------------------------------------------------
Sentence: Microsoft acquired Activision Blizzard for $69 billion.
Word: Microsoft | Label: I-ORG | Score: 0.9995999932289124
Word: Act | Label: I-ORG | Score: 0.9994999766349792
Word: ##ivision | Label: I-ORG | Score: 0.9983000159263611
Word: B | Label: I-ORG | Score: 0.9994000196456909
Word: ##lizzard | Label: I-ORG | Score: 0.9979000091552734
--------------------------------------------------
Sentence: The Chandrayaan-3 mission was successfully launched by ISRO.
Word: Chandra | Label: I-MISC | Score: 0.9972000122070312
Word: ##ya | Label: I-MISC | Score: 0.9815999865531921
Word: ##an | Label: I-MISC | Score: 0.996999979019165
Word: - | Label: I-MISC | Score: 0.9825000166893005
Word: 3 | Label: I-MISC | Score: 0.9898999929428101
Word: IS | Label: I-ORG | Score: 0.9976000189781189
Word: ##RO | Label: I-ORG | Score: 0.996999979019165
--------------------------------------------------
```

STEP 10 — Clean Transformer Output

```
hf_results_clean = []

for sentence in sentences:
    results = ner_pipeline(sentence)
    for r in results:
        word = r['word'].replace("##", "")
        hf_results_clean.append([
            sentence,
            word,
            r['entity'],
            round(r['score'],4)
        ])
```

```
df_hf = pd.DataFrame(hf_results_clean,
                     columns=["Sentence", "Entity", "Label", "Confidence Score"])

df_hf
```

	Sentence	Entity	Label	Confidence Score
0	Apple announced new AI features at WWDC 2024 i...	Apple	I-ORG	0.9978
1	Apple announced new AI features at WWDC 2024 i...	W	I-MISC	0.9348
2	Apple announced new AI features at WWDC 2024 i...	WD	I-MISC	0.9257
3	Apple announced new AI features at WWDC 2024 i...	C	I-MISC	0.9882
4	Apple announced new AI features at WWDC 2024 i...	California	I-LOC	0.9996
5	Virat Kohli scored a century in the match agai...	V	I-PER	0.9996
6	Virat Kohli scored a century in the match agai...	ira	I-PER	0.9963
7	Virat Kohli scored a century in the match agai...	t	I-PER	0.9991
8	Virat Kohli scored a century in the match agai...	Ko	I-PER	0.9995
9	Virat Kohli scored a century in the match agai...	hl	I-PER	0.9907
10	Virat Kohli scored a century in the match agai...	i	I-PER	0.9947
11	Virat Kohli scored a century in the match agai...	Australia	I-LOC	0.9997
12	Virat Kohli scored a century in the match agai...	Mumbai	I-LOC	0.9990
13	Prime Minister Narendra Modi visited the Unite...	Na	I-PER	0.9992
14	Prime Minister Narendra Modi visited the Unite...	ren	I-PER	0.9925
15	Prime Minister Narendra Modi visited the Unite...	dra	I-PER	0.9984
16	Prime Minister Narendra Modi visited the Unite...	Mo	I-PER	0.9989
17	Prime Minister Narendra Modi visited the Unite...	di	I-PER	0.9838
18	Prime Minister Narendra Modi visited the Unite...	United	I-LOC	0.9997
19	Prime Minister Narendra Modi visited the Unite...	States	I-LOC	0.9998
20	Microsoft acquired Activision Blizzard for $69...	Microsoft	I-ORG	0.9996
21	Microsoft acquired Activision Blizzard for $69...	Act	I-ORG	0.9995
22	Microsoft acquired Activision Blizzard for $69...	ivision	I-ORG	0.9983
23	Microsoft acquired Activision Blizzard for $69...	B	I-ORG	0.9994
24	Microsoft acquired Activision Blizzard for $69...	lizzard	I-ORG	0.9979
25	The Chandrayaan-3 mission was successfully lau...	Chandra	I-MISC	0.9972
26	The Chandrayaan-3 mission was successfully lau...	ya	I-MISC	0.9816
27	The Chandrayaan-3 mission was successfully lau...	an	I-MISC	0.9970
28	The Chandrayaan-3 mission was successfully lau...	-	I-MISC	0.9825
29	The Chandrayaan-3 mission was successfully lau...	3	I-MISC	0.9899
30	The Chandrayaan-3 mission was successfully lau...	IS	I-ORG	0.9976
31	The Chandrayaan-3 mission was successfully lau...	RO	I-ORG	0.9970

STEP 11 — Comparison Table

Feature	spaCy	Hugging Face
Model Type	Statistical model	Transformer (BERT)
Speed	Faster	Slower
Accuracy	Moderate	Higher
Context Handling	Limited	Strong contextual embeddings
Confidence Score	No	Yes
GPU Requirement	No	Recommended

STEP 12 — Result Analysis (8–10 Sentences)

In this experiment, both spaCy and Transformer-based NER models were applied to real-world sentences. The Transformer model detected slightly more entities and provided confidence scores, which spaCy does not provide by default. The Transformer model handled context better, especially for complex entity names. spaCy performed faster and required less computational resources. Transformer models use contextual embeddings, which allow them to understand word meaning based on surrounding words. However,

they require more memory and often benefit from GPU acceleration. spaCy is more suitable for real-time applications due to its speed. Transformers are better suited for research or high-accuracy tasks. Overall, the Transformer model achieved better qualitative performance.

STEP 13 — Error Analysis

❌ Error 1

Sentence: "Apple announced new AI features at WWDC 2024 in California."

Issue: spaCy may label Apple as ORG correctly but sometimes miss WWDC.

Reason: WWDC is a specific tech event and may not appear frequently in training data.

❌ Error 2

Sentence: "Virat Kohli scored a century in Mumbai."

Issue: Transformer may split “Virat” and “Kohli” into subwords.

Reason: BERT tokenization (WordPiece) splits rare words.

❌ Error 3

Sentence: "The Chandrayaan-3 mission was successfully launched by ISRO."

Issue: Model may misclassify Chandrayaan-3.

Reason: Hyphenated scientific names are harder to classify.

```
Start coding or generate with AI.
```